1.
    a. if (grade >= 90)
    b. if (number < 20 || number > 50)
    c. if (y < 100)
2. if (num1 > num2) {
        System.out.println(num1 + " is greater than " + num2);
   } else if (num2 > num1) {

        System.out.println(num2 + " is greater than " + num1);

   } else {

        System.out.println(num1 + " and " + num2 + " are equal.");

   }

3.
    a. even, odd
    b. switch (num % 2) {

        case 0:

            System.out.println("Even")
            break;

        case Default:

            System.out.println("Odd")
            break;

        }

4.
    a. random.nextInt(1, 50 + 1);
    b. random.nextInt(20, 100 + 1);
    c. random.nextdouble(10, 20 + 1);

5. The decision statements do not catch the exact values specified. None of the conditionals catch 18 or 65.

6.
    a. True
    b. False
    c. True

d. True
e. True
f. True
g. True

7.

a. Math.pow(y, x);
b. Math.abs(y);
c. Math.sqrt(y);

8.

a. True
b. False, integers cannot contain decimals so a roundoff error cannot happen.
c. False, the nested if statement runs if the statement outside is true. The if-else statement runs if the statement before is false.
d. False, the expression can be a string, char, enum, int, short, etc.
e. True.
f. True.
g. True.
h. True.
i. True.
j. True.
k. True.